Infant Formula Heavy-Metal Standard

Method v3.0 — the strictest government limit in the world, for every analyte

Institute of Contaminant Standards
Operator of the Heavy Metal Tested & Certified program
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The Heavy Metal Tested & Certified (HMTc) program certifies infant formula against **the single strictest maximum level set by any government in the world** for each heavy metal, in the product's own basis. Where two governments regulate the same metal, the tighter number binds; where the strictest limit is set as a ready-to-feed (as-consumed) value, it is converted to powder basis across the 1:7 reconstitution ratio before comparison, and vice versa. The certification limit is therefore never looser than any national law, and in most cells it is tighter than the law a given brand's home market would apply.

This document shows every government's limit, per subcategory, so that a brand, a retailer, a regulator, or opposing counsel can check the derivation directly. Each certification number is the lowest comparable value in its row; nothing is asserted that a sovereign government has not already put in force. The prior occurrence-percentile methodology is retired: it repeatedly produced defects (basis mis-conversions, poisoned pools, dead branches), and a lookup against sovereign law is both faster and unimpeachable.

The standard — one number per analyte × subcategory (µg/kg)

Subcat-egory	Pb	iAs	Cd	Hg	Ni	Al	Sn	Cr
Powder · cow-milk	20	20	10	mon.	250	3500	35000	2000[d]

Subcategory	Pb	iAs	Cd	Hg	Ni	Al	Sn	Cr
Powder · soy	20	20	20	mon.	400	7000	35000	2000[d]
Ready-to-feed · cow-milk	10	10	5	mon.	100	500	5000	mon.
Ready-to-feed · soy	10	10	10	mon.	100	1000	5000	mon.

mon. = monitored, not limited (no government sets a limit; reflex speciation screen applies). [d] *= broad-dairy fallback, no formula-specific limit. Powder rows are powder-as-sold basis; ready-to-feed rows are as-consumed basis. Every value is the strictest government maximum level worldwide for that product form; the derivation is the per-country matrix below.*

How certification works against these limits

Tier 1 metals — lead, inorganic arsenic, cadmium, mercury — are zero-tolerance. A single lot over the limit blocks certification. These are the metals with no safe intake for infants, so the program holds them to the strictest world limit with no transitional allowance.

Tier 2 metals — nickel, aluminium, tin, hexavalent chromium — carry a transitional allowance. A product measuring up to **150 % of the limit** earns transitional certification (Status B) while it remediates, because these metals have established tolerable intakes and their elevations are often supply-chain-driven rather than avoidable at will. Above 150 % is probation; the allowance is transitional, not permanent.

A legal violation always blocks certification, regardless of HMTc status. Because every limit here is a government maximum level, any lot that fails the *program* limit by more than the Tier 2 allowance is at or near a real legal exceedance in the strictest jurisdiction — the certification gate and the regulatory gate move together.

Certification status	Meaning
A — Full	All analytes ≤ 100 % of limit
B — Transitional	A Tier 2 metal between 100–150 % of limit, under a remediation plan
C — Probation	150–200 %, or any Tier 1 exceedance
D — Suspension	> 200 % of limit
E — Revocation	Repeated or unremediated failure

Per-country government limits, by subcategory

CX = Codex · CA = Canada · CN = China · IN = India · BR = Brazil · KR = South Korea · GS = Gulf (GSO) · AN = Australia/NZ · US = US-FDA.

Each cell is that government's infant-formula limit **normalized to the subcategory's basis** so the columns are comparable. [c] = converted across the 1:7 powder↔as-consumed reconstitution boundary (the government's own limit is in the other basis; see the provenance table). Blank = that government sets no formula limit for the analyte. **The HMTc limit is the lowest comparable value in the row.**

Powder · cow-milk (powder, as-sold basis, µg/kg)

Analyte	EU	UK	CX	CA	CN	IN	BR	GS	AN	HMTc limit	Binding source
Lead (Pb)	20	50	70[c]	70[c]	80	140[c]	70[c]	70[c]	70[c]	**20**	EU
Inorganic arsenic (iAs)	20						140[c]			**20**	EU
Cadmium (Cd)	10	10				700[c]	70[c]			**10**	EU
Mercury (tHg)										—	monitored *(see note)*
Nickel (Ni)	250									**250**	EU
Aluminium (Al)									3500[c]	**3500**	AN (500 as-consumed ×7)
Tin (Sn)						35000[c]				**35000**	IN (5000 as-consumed ×7)
Total chromium (Cr)					2000[d]					**2000**	CN *(broad-dairy fallback)*

Canned formula tin (separate product form): EU 50000, UK 50000 µg/kg. [d] *= broad-dairy fallback (no formula-specific limit exists).* [c] *= basis-converted.*

Powder · soy (powder, as-sold basis, µg/kg)

Analyte	EU	UK	CX	CA	CN	IN	BR	AN	HMTc limit	Binding source
Lead (Pb)	20	50	70[c]	70[c]	80	140[c]	70[c]	70[c]	**20**	EU
Inorganic arsenic (iAs)	20						140[c]		**20**	EU
Cadmium (Cd)	20	20				700[c]	70[c]		**20**	EU
Mercury (tHg)									—	monitored *(see note)*
Nickel (Ni)	400								**400**	EU
Aluminium (Al)								7000[c]	**7000**	AN (1000 as-consumed ×7)
Tin (Sn)						35000[c]			**35000**	IN (5000 as-consumed ×7)
Total chromium (Cr)					2000[d]				**2000**	CN *(broad-dairy fallback)*

Canned soy formula tin (separate product form): EU 50000, UK 50000 µg/kg. [d] *= broad-dairy fallback (no formula-specific limit exists).* [c] *= basis-converted.*

Ready-to-feed · cow-milk (as-consumed basis, µg/kg)

Analyte	EU	UK	CX	CA	CN	IN	BR	GS	AN	HMTc limit	Binding source
Lead (Pb)	10	10	10	10	10	20	10	10	10	**10**	EU

Analyte	EU	UK	CX	CA	CN	IN	BR	GS	AN	HMTc limit	Binding source
Inorganic arsenic (iAs)	10						20			**10**	EU
Cadmium (Cd)	5	5				100	10			**5**	EU
Mercury (tHg)										—	monitored *(see note)*
Nickel (Ni)	100									**100**	EU
Aluminium (Al)									500	**500**	AN
Tin (Sn)						5000				**5000**	IN
Total chromium (Cr)										—	monitored *(see note)*

Canned formula tin (separate product form): EU 50000, UK 50000 µg/kg. [d] *= broad-dairy fallback (no formula-specific limit exists).* [c] *= basis-converted.*

Ready-to-feed · soy (as-consumed basis, µg/kg)

Analyte	EU	UK	CX	CA	CN	IN	BR	AN	HMTc limit	Binding source
Lead (Pb)	10	10	10	10	10	20	10	10	**10**	EU
Inorganic arsenic (iAs)	10						20		**10**	EU
Cadmium (Cd)	10	10				100	10		**10**	EU
Mercury (tHg)									—	monitored *(see note)*

Ana-lyte	EU	UK	CX	CA	CN	IN	BR	AN	HMTc limit	Bind-ing source
Nickel (Ni)	100								**100**	EU
Aluminium (Al)								1000	**1000**	AN
Tin (Sn)						5000			**5000**	IN
Total chromium (Cr)									—	moni-tored *(see note)*

Canned soy formula tin (separate product form): EU 50000, UK 50000 µg/kg. [d] *= broad-dairy fallback (no formula-specific limit exists).* [c] *= basis-converted.*

Three cells where the method reaches its limit – stated plainly

The tightest-government method is only as complete as the world's regulators. Three cells cannot be set by lookup, and this standard says so rather than inventing a number:

1. **Mercury (all subcategories) — no government sets an infant-formula limit.** The nearest limit in the world is China's 20 µg/kg for *canned complementary foods*, a different product. There is nothing to adopt. Mercury is therefore **monitored, not limited**: every certification lot is screened for total mercury, with reflex methylmercury speciation on any detection, but no numeric pass/fail limit is published where no government has set one. This is the honest output of the method, not an omission.
2. **Hexavalent chromium — unregulated in food worldwide.** No government sets a Cr-VI food limit (the only Cr-VI limits are for drinking water). Total chromium has one applicable government number — China's 2,000 µg/kg for milk powder (a broad dairy category that covers cow-milk formula powder), shown with a [d] flag as a fallback, not a formula-specific limit. Cr-VI itself is **monitored, not limited**, with a 100 µg/kg total-chromium manufacturing-anomaly screen that triggers speciation.
3. **Tin — regulated, but on an uncertain basis.** Only India sets a non-canned infant-formula tin limit (5,000 µg/kg); every other government regulates tin only for *canned* product (50,000 µg/kg, a container-migration limit). The India value's basis is unresolved in our register (the secondary source records it as-consumed; the FSSAI 2011 primary text records it as-sold — a 7× difference). The tin cell is therefore **provisional** pending confirmation against the primary Indian instrument; canned formula is held to the 50,000 µg/kg world consensus.

Everything else in this standard is a firm, citable government maximum level.

Provenance — every limit exactly as its government states it

Analyte	Gov	Limit (µg/kg)	Basis (as stated)	Product scope	Source
Pb	AU-NZ	10	portion ordinarily consumed	infant formula products	regulations/au-nz-food-standards-code-sc
Pb	BR	10	as consumed	Pediatric formula for enteral nutrition (fórmula pediátrica para nutri	regulations/br-infant-food-contaminants
Pb	BR	10	as consumed	Nutrient formulas for high-risk newborns (fórmulas de nutrientes para	regulations/br-infant-food-contaminants
Pb	BR	10	as consumed	Follow-up / follow-on infant formulas (fórmulas infantis de seguimento	regulations/br-infant-food-contaminants
Pb	BR	10	as consumed	Infant formulas for special dietary/dietotherapeutic purposes (necessi	regulations/br-infant-food-contaminants
Pb	BR	10	as consumed	Infant formulas for infants / starter (fórmulas infantis para lactente	regulations/br-infant-food-contaminants
Pb	BR	10	as consumed	Other foods specially formulated for infants and young children (outro	regulations/br-infant-food-contaminants
Pb	CA	10	as consumed	Infant formula (all types); two former limits — concen-	regulations/ca-infant-food-contaminants

Analyte	Gov	Limit (µg/kg)	Basis (as stated)	Product scope	Source
				trated/powdered	
Pb	CN	80	product as placed on market	Formula food for infants and young children (powdered infant/follow-up	regulations/cn-infant-food-contaminants
Pb	CN	10	as consumed	Liquid (ready-to-feed) infant formula	regulations/cn-infant-food-contaminants
Pb	CN	150	product as placed on market	Formulas for special medical purposes (FSMP), products for children 1-	regulations/cn-infant-food-contaminants
Pb	CN	500	product as placed on market	Formulas for special medical purposes (FSMP), products for children ov	regulations/cn-infant-food-contaminants
Pb	CN	150	product as placed on market	Formulas for special medical purposes (excl. infant FSMP), products fo	regulations/cn-gb2762-2022
Pb	CODEX	10	as consumed	Infant formula, formula for special medical purposes intended for infa	regulations/codex-infant-food-contaminan
Pb	EU	20	product as placed on market	infant formulae, follow-on formulae, and young-child formulae placed o	regulations/eu2023-contaminants-maximum-
Pb	EU	10	product as placed on market	infant formulae, follow-on formulae, and young-	regulations/eu2023-contaminants-maximum-

Analyte	Gov	Limit (µg/kg)	Basis (as stated)	Product scope	Source
				child formulae placed o	
Pb	EU	20	ready to use	drinks for infants and young children placed on the market and labelle	regulations/eu2023-contaminants-maximum-
Pb	GSO	10	as consumed	Infant formula, formula for special medical purposes intended for infa	regulations/gso-infant-food-contaminants
Pb	IN	200	as consumed	Infant Milk substitute and Infant foods	regulations/in-infant-food-contaminants
Pb	IN	20	as consumed	Infant formula (ready to use)	regulations/in-infant-food-contaminants
Pb	UK	50	product as placed on market	Infant formulae and follow-on formulae, marketed as powder	regulations/uk-infant-food-contaminants
Pb	UK	10	as consumed	Infant formulae and follow-on formulae, marketed as liquid (ready-to-f	regulations/uk-infant-food-contaminants
iAs	BR	20	as consumed	Pediatric formula for enteral nutrition (fórmula pediátrica para nutri	regulations/br-infant-food-contaminants
iAs	BR	20	as consumed	Nutrient formulas for high-risk newborns (fórmulas de nutrientes para	regulations/br-infant-food-contaminants
iAs	BR	20	as consumed	Follow-up / follow-on infant formu-	regulations/br-infant-food-contaminants

Analyte	Gov	Limit (µg/kg)	Basis (as stated)	Product scope	Source
				las (fórmulas infantis de seguimento	
iAs	BR	20	as consumed	Infant formulas for special dietary/dietotherapeutic purposes (necessi	regulations/br-infant-food-contaminants
iAs	BR	20	as consumed	Infant formulas for infants / starter (fórmulas infantis para lactente	regulations/br-infant-food-contaminants
iAs	BR	20	as consumed	Other foods specially formulated for infants and young children (outro	regulations/br-infant-food-contaminants
iAs	EU	20	product as placed on market	infant formulae, follow-on formulae, food for special medical purposes	regulations/eu2023-contaminants-maximum-
iAs	EU	10	product as placed on market	infant formulae, follow-on formulae, food for special medical purposes	regulations/eu2023-contaminants-maximum-
tAs	CN	500	product as placed on market	Milk powder and modified milk powder (general dairy category, distinct	regulations/cn-infant-food-contaminants
tAs	IN	50	as consumed	Infant Milk substitute and Infant foods	regulations/in-infant-food-contaminants
Cd	BR	10	as consumed	Pediatric formula for enteral nutrition (fórmula	regulations/br-infant-food-contaminants

Analyte	Gov	Limit (µg/kg)	Basis (as stated)	Product scope	Source
				pediátrica para nutri	
Cd	BR	10	as consumed	Nutrient formulas for high-risk newborns (fórmulas de nutrientes para	regulations/br-infant-food-contaminants
Cd	BR	10	as consumed	Follow-up / follow-on infant formulas (fórmulas infantis de seguimento	regulations/br-infant-food-contaminants
Cd	BR	10	as consumed	Infant formulas for special dietary/dietotherapeutic purposes (necessi	regulations/br-infant-food-contaminants
Cd	BR	10	as consumed	Infant formulas for infants / starter (fórmulas infantis para lactente	regulations/br-infant-food-contaminants
Cd	BR	10	as consumed	Other foods specially formulated for infants and young children (outro	regulations/br-infant-food-contaminants
Cd	EU	10	product as placed on market	infant formulae, follow-on formulae, food for special medical purposes	regulations/eu-2023-915-cadmium
Cd	EU	5	product as placed on market	infant formulae, follow-on formulae, food for special medical purposes	regulations/eu-2023-915-cadmium

Analyte	Gov	Limit (µg/kg)	Basis (as stated)	Product scope	Source
Cd	EU	20	product as placed on market	infant formulae, follow-on formulae, food for special medical purposes	regulations/eu-2023-915-cadmium
Cd	EU	10	product as placed on market	infant formulae, follow-on formulae, food for special medical purposes	regulations/eu-2023-915-cadmium
Cd	EU	20	ready to use	drinks for infants and young children labelled and placed on the marke	regulations/eu-2023-915-cadmium
Cd	IN	100	as consumed	Infant Milk substitute and Infant foods	regulations/in-infant-food-contaminants
Cd	UK	10	product as placed on market	Infant formulae and follow-on formulae, powdered formulae manufactured	regulations/uk-infant-food-contaminants
Cd	UK	5	product as placed on market	Infant formulae and follow-on formulae, liquid formulae manufactured f	regulations/uk-infant-food-contaminants
Cd	UK	20	product as placed on market	Infant formulae and follow-on formulae, powdered formulae manufactured	regulations/uk-infant-food-contaminants
Cd	UK	10	product as placed on market	Infant formulae and follow-on formulae, liquid formulae manufactured f	regulations/uk-infant-food-contaminants

Analyte	Gov	Limit (µg/kg)	Basis (as stated)	Product scope	Source
Ni	EU	250	product as placed on market	infant/young-child formula, powder, except soy-isolate	regulations/eu-2024-1987-nickel-maximum-
Ni	EU	400	product as placed on market	infant/young-child formula, powder, soy protein isolate	regulations/eu-2024-1987-nickel-maximum-
Ni	EU	100	product as placed on market	infant/young-child formula, liquid	regulations/eu-2024-1987-nickel-maximum-
Al	AU-NZ	500	portion ordinarily consumed	infant formula, follow-on formula, and special medical purpose product	regulations/au-nz-food-standards-code-sc
Al	AU-NZ	1000	portion ordinarily consumed	soy-based infant formula products	regulations/au-nz-food-standards-code-sc
Sn	EU	50000	product as placed on market	canned infant formulae, canned follow-on formulae, and canned young-ch	regulations/eu2023-contaminants-maximum-
Sn	IN	5000	as consumed	Infant Milk substitute and Infant foods	regulations/in-infant-food-contaminants
Sn	UK	50000	as consumed	Canned infant formulae and follow-on formulae, including infant milk a	regulations/uk-infant-food-contaminants
Cr-total	CN	2000	product as placed on market	Milk powder and modified milk powder (general dairy category, distinct	regulations/cn-infant-food-contaminants

Provenance and review status

Every value above is drawn from `data/evidence/regulatory_limits.csv` (250-row worldwide regulatory register). China, Brazil, India, US-FDA, and Codex infant-food rows are confirmed against primary government instruments (marked PRIMARY-CONFIRMED in the register, 2026-08-03); the remaining jurisdictions (EU, UK, Australia/NZ, Canada, Gulf, South Korea) are web-verified against the official instrument and pending a primary-text cross-check. The two provisional cells above (India tin basis; the Cr-VI/mercury monitored-not-limited determinations) and the primary-text cross-check on the remaining jurisdictions are the items outstanding before external publication. **This standard is ready for legal-counsel review; it is not yet counsel-cleared.**

The Institute of Contaminant Standards operates the Heavy Metal Tested & Certified program. This is certification-program material, published separately from the Heavy Metal Index literature reference at heavymetalindex.com.